

Math 421: Homework 2

Solutions

Due: Sept 19, 2025

Problem 1: Determine whether the following statements are true or false and provide a short justification -

A. $\{\{\emptyset\}\} \cup \emptyset = \{\emptyset, \{\emptyset\}\}$

FALSE. As the empty set has no elements, the union is just $\{\{\emptyset\}\}$.

B. $\{\{\emptyset\}\} \cup \{\emptyset\} = \{\emptyset, \{\emptyset\}\}$

TRUE. The first set has one element, $\{\emptyset\} \in \{\{\emptyset\}\}$, and the second set also has one element, $\emptyset \in \{\emptyset\}$, so the union consists of both elements $\{\emptyset, \{\emptyset\}\}$.

C. $\{\emptyset, \{\emptyset\}\} \cap \{\{\emptyset\}, \{\{\emptyset\}\}\} = \{\emptyset\}$

FALSE. $\emptyset \notin \{\{\emptyset\}, \{\{\emptyset\}\}\}$.

D. $\{\emptyset, \{\emptyset\}\} \cap \{\{\emptyset\}, \{\{\emptyset\}\}\} = \{\{\emptyset\}\}$

TRUE. $\{\emptyset\} \in \{\emptyset, \{\emptyset\}\}$ and $\{\emptyset\} \in \{\{\emptyset\}, \{\{\emptyset\}\}\}$.

Problem 2: (Ross Chapter 1 1.11) For each $n \in \mathbb{N}$, let P_n denote the assertion “ $n^2 + 5n + 1$ is an even integer.”

A. Prove P_{n+1} is true whenever P_n is true.

Proof: Assume P_n is true, that is, $n^2 + 5n + 1$ is an even integer. Then there exists $k \in \mathbb{Z}$ such that $n^2 + 5n + 1 = 2k$. Consider

$$(n+1)^2 + 5(n+1) + 1 = n^2 + 2n + 1 + 5n + 5 + 1 = (n^2 + 5n + 1) + 2n + 6 = 2k + 2n + 6 = 2(k + n + 3).$$

Thus $(n+1)^2 + 5(n+1) + 1$ is an even integer and P_{n+1} is true whenever P_n is true.

B. P_n is false for every n , since if n is even, $n = 2k$ for some $k \in \mathbb{Z}$ and

$$(2k)^2 + 5(2k) + 1 = 2(2k^2 + 5k) + 1$$

an odd integer. If n is odd, $n = 2k + 1$ for some $k \in \mathbb{Z}$ and

$$(2k+1)^2 + 5(2k+1) + 1 = 4k^2 + 4k + 1 + 10k + 5 + 1 = 2(2k^2 + 7k + 3k) + 1$$

and odd integer. Notice that we did not check the base case, P_1 , had we done so, we would see

$$1^2 + 5 \cdot 1 + 1 = 7$$

which is not even, so the base case does not hold.

Problem 3: Prove using induction: If x is a real number and $x \neq 1$, then

$$\sum_{i=0}^n x^i = \frac{1 - x^{n+1}}{1 - x}$$

for every $n \in \mathbb{N}$.

Proof: Let P_n be the assertion “For $x \in \mathbb{R}$ and $x \neq 1$, $\sum_{i=0}^n x^i = \frac{1 - x^{n+1}}{1 - x}$.”

Base Case: Let $n = 1$, then P_1 is true as

$$\sum_{i=0}^1 x^i = x^0 + x^1 = 1 + x$$

and

$$\frac{1 - x^{1+1}}{1 - x} = 1 + x.$$

Inductive Step: Assume P_n is true for some $n = k$. That is, for $x \in \mathbb{R}$ and $x \neq 1$, $\sum_{i=0}^k x^i = \frac{1-x^{k+1}}{1-x}$. then,

$$\begin{aligned} \sum_{i=0}^{k+1} x^i &= \sum_{i=0}^k x^i + x^{k+1} = \frac{1-x^{k+1}}{1-x} + x^{k+1} \\ &= \frac{1-x^{k+1}}{1-x} + \frac{(1-x)(x^{k+1})}{1-x} \\ &= \frac{1-x^{k+1} + x^{k+1} - x^{k+2}}{1-x} \\ &= \frac{1-x^{k+2}}{1-x}. \end{aligned}$$

Therefore P_n is true for $n = k + 1$ whenever it is true for $n = k$, and the assertion holds for all $n \in \mathbb{N}$ by the principle of mathematical induction.

Problem 4: (Ross Chapter 1 1.9)

(a) $2^n > n^2$ is true for $n = 0, 1$ and for all integers $n \geq 5$.

(b) For $n \geq 5$, let P_n be the assertion that $2^n > n^2$.

Base Case: P_5 is true as $32 = 2^5 < 5^2 = 25$.

Inductive Step: Assume P_n is true for $n \geq 5$. Since $n \geq 5$, we have

$$(n-1)^2 > (5-1)^2 > 4^2 > 2$$

so that

$$n^2 - 2n + 1 > 2 \Rightarrow n^2 - 2n - 1 > 0 \Rightarrow n^2 > 2n + 1.$$

Therefore

$$2^{n+1} = 2 \cdot 2^n > 2n^2 = n^2 + n^2 > n^2 + 2n + 1 = (n+1)^2$$

and P_{n+1} holds if P_n holds. Therefore P_n holds for all $n \geq 5$ by the principle of mathematical induction.

Problem 5: Prove the following theorem.

Theorem: Suppose A and B are subsets of a set S . If $A \subseteq B$, then $B^c \subseteq A^c$.

Proof. Let A and B be subsets of a set S and suppose $A \subseteq B$. Let $x \in B^c$, then $x \notin B$. But, since every element of A is also in B , $x \notin A$, and therefore $x \in A^c$. Thus $B^c \subseteq A^c$.

Problem 6: Prove, using a proof by contradiction, that

$$\{2k+1 : k \in \mathbb{N}\} \cap \{4k : k \in \mathbb{N}\} = \emptyset.$$

Proof. Assume by way of contradiction that $\{2k+1 : k \in \mathbb{N}\} \cap \{4k : k \in \mathbb{N}\} \neq \emptyset$. That is, there exists $x \in \{2k+1 : k \in \mathbb{N}\} \cap \{4k : k \in \mathbb{N}\}$. By definition of intersection, this means that $x \in \{2k+1 : k \in \mathbb{N}\}$ and $x \in \{4k : k \in \mathbb{N}\}$. Thus, there exists a $k_1 \in \mathbb{N}$ such that $x = 2k_1 + 1$, and therefore x is an odd integer, and there exists a $k_2 \in \mathbb{N}$ such that $x = 4k_2 = 2(2k_2)$ and therefore x is an even integer. But this is a contradiction as an integer cannot both be even and odd. Thus our assumption that $\{2k+1 : k \in \mathbb{N}\} \cap \{4k : k \in \mathbb{N}\} \neq \emptyset$ must be false and therefore

$$\{2k+1 : k \in \mathbb{N}\} \cap \{4k : k \in \mathbb{N}\} = \emptyset.$$